

Bias Busters, A Comprehensive Approach to Fairness in News-Based NLP Models

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Bias Detection System

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Summary of Stage-1:

Data Collection:

- Gathered labeled news articles from Kaggle and OpenSources.
- Created dataset of potentially biased and neutral content.
- Ensured diversity across bias categories .

Initial Approach:

- Applied unsupervised clustering (K-means) to identify bias patterns.
- Used TF-IDF vectorization to extract important features from text.
- Established baseline classification between biased and unbiased content.
- Achieved initial categorization without supervised labels.

Summary of Stage-2:

Enhanced Detection:

- Moved from unsupervised to supervised learning models.
- Implemented Logistic Regression classifier for better accuracy.
- Added image analysis capabilities for representation bias.

Mitigation Strategies:

- Developed rule-based term replacement system.
- Created context-aware text rewriting using T5 language model.
- Provided actionable recommendations for inclusive representation in images.

User Interface:

- Built intuitive Gradio interface for easy interaction.
- Created separate analysis paths for text and image content.

Abstract

The rise of biased news content poses a threat to media integrity and public trust. This project aims to create a system that detects bias in news articles using natural language processing, machine learning, and data analytics. Our approach focuses on identifying bias indicators, such as sensationalist language and framing, to assess the fairness of news content. Various machine learning algorithms are employed to classify articles based on their bias levels, helping to reveal elements contributing to bias. This project aims to promote fair media practices and offer a tool for journalists, researchers, and readers.

Problem Statement

- Media bias in news articles can distort facts, shape public opinion, and reduce trust in journalism.
- The goal is to develop a system that uses Natural Language Processing (NLP) and machine learning techniques to automatically detect and classify biases in news articles.
- This system aims to improve the accuracy of bias detection and help mitigate the impact of biased reporting on public understanding.

- Develop an algorithm to detect bias in news content using linguistic and contextual features.
- Build a machine learning model to classify bias levels across diverse news sources.
- Provide analytics that highlight key bias indicators in articles to support transparency.
- Generate context-aware, neutral alternatives for biased content using advanced language models.
- Design a user-friendly platform accessible to journalists, researchers, and general readers.

Base Paper

Title	Dbias: Detecting Biases and Ensuring Fairness in News Articles
Source	International Journal of Data Science and Analytics
DOI	https://doi.org/10.1007/s41060-022-00359-4
Published	01 September 2022
Authors	Shaina Raza, Deepak John Reji, Chen Ding
Technologies	NLP and Machine Learning Deep Neural Networks Transformer Models (e.g., BERT, DistilBERT)
Algorithms	Bias Detection Module Bias Recognition Module De-biasing via Masking and Infill
Objective	A fair ML pipeline for detecting, recognizing, and mitigating biases

Literature Review

Table: Summary of Reviewed Literature

S.No	Title	Authors	Year & Dataset	Methodology	Results	Remarks
1	Developing a Natural Language Understanding Model to Characterize Cable News Bias	Seth P. Benson, Iain J. Cruickshank	2024, Transcripts from Fox, MSNBC, CNN (14,000 samples)	Named Entity Recognition (NER) for topic selection, Stance Analysis with GPT-4	Identified consistent bias clusters across networks	Shows potential for objective bias detection in cable news
2	The Effects of Media Bias on News Recommendations	Qin Ruan, Brian Mac Namee, Ruihai Dong	2024, MIND and SemEval-2019 datasets	Comparison of state-of-the-art news recommendation algorithms for bias propagation	Highlighted algorithms' tendencies to amplify user biases	Indicates need for mitigation in personalized news recommendation systems
3	Dbias: Detecting Biases and Ensuring Fairness in News Articles	Shaina Raza, Deepak John Reji, Chen Ding	2022, Small text dataset for training	Bias detection using deep learning and masking strategies	Demonstrated bias removal via masking and infilling techniques	Effective for small datasets but less scalable

Literature Review

Table: Summary of Reviewed Literature

S.No	Title	Authors	Year & Dataset	Methodology	Results	Remarks
4	Deep Learning for Fake News Detection on Social Media: A Survey	R. S. P. Gupta, S. G. N. M. Kumar	2023, Public News Dataset	Overview of deep learning models for fake news detection	Discussed various DL approaches, performance metrics, and challenges	Highlights importance of using diverse datasets for improving model accuracy
5	Enhancing News Recommendation Systems: Challenges and Solutions	S. Ghosh, P. Sharma	2023, Large News Dataset	Comparative analysis of content-based and collaborative filtering techniques	Demonstrated the effectiveness of hybrid approaches in improving accuracy	Suggests incorporating personalization for better user engagement in recommendations
6	A Framework for Bias Detection in News Articles: Approaches and Future Directions	M. V. Gupta, L. T. R. Sharma	2022, News Dataset (100,000 articles)	Bias detection using NLP models and adversarial training	Found that adversarial models outperformed traditional ones in detecting subtle biases	Emphasizes the need for adversarial training in mitigating hidden biases in news content

Dataset Analysis and Bias Details

- **Dataset Overview:**

- **Number of Columns:** 1
- **Number of Rows:** 178
- **Total Number of Words:** 6,097

- **Data Splitting:**

- **Training Data:** 80%
- **Testing Data:** 20%

Methodology

Data Preprocessing:

- Performed text normalization and cleaning (lowercasing, removal of special characters).
- Removed stopwords using NLTK.
- Applied tokenization for feature extraction.

Feature Engineering:

- Used TF-IDF vectorization with 1-2 n-grams.
- Extracted most significant features.
- Generated cluster-based labels to support supervised learning.

Model Development:

- Applied K-means clustering to separate biased and unbiased content.
- Trained Logistic Regression model with an 80/20 train-test split.
- Fine-tuned language models for context-aware biased text rewriting.

Implementation Details

Tools and Frameworks Used:

- Python for data processing and model development.
- `scikit-learn` for ML algorithms (K-means, Logistic Regression).
- NLTK for natural language processing tasks.
- Transformers library (BERT, T5) for advanced NLP capabilities.
- spaCy for linguistic feature extraction.
- Gradio for building the interactive user interface.
- Pandas for data manipulation and analysis.

System Components:

- Text preprocessing module.
- Bias detection classifiers.
- Term-based and context-based mitigation engines.
- Image representation analyzer.
- Interactive web interface.

Model Training Code

```
from sklearn.linear_model import LogisticRegression

# Initialize and train the logistic regression model
model = LogisticRegression()
model.fit(X_train, y_train)

# Make predictions on the test set
y_pred = model.predict(X_test)

# Evaluate the model
from sklearn.metrics import accuracy_score, classification_report

# Calculate and print accuracy and classification report
print(f"Accuracy: {accuracy_score(y_test, y_pred):.2f}")
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

Accuracy: 0.81

Classification Report:

	precision	recall	f1-score	support
0	0.81	1.00	0.89	29
1	0.00	0.00	0.00	7
accuracy			0.81	36
macro avg	0.40	0.50	0.45	36
weighted avg	0.65	0.81	0.72	36

```
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

Model Training Code

```
from sklearn.metrics import accuracy_score

# Ensure that val_labels and val_preds are on the CPU
val_labels = val_labels.cpu() # If val_labels is a tensor, move it to CPU

# Get the text data from the validation set
val_texts = val_data['text'].tolist() # Assuming 'text' is the column name for your text data

# Generate predictions for the validation set
val_preds = []
for text in val_texts:
    inputs = tokenizer(text, return_tensors="pt", padding=True, truncation=True, max_length=128)
    logits = model(**inputs).logits
    predicted_class = torch.argmax(logits, dim=-1).item()
    val_preds.append(predicted_class)

# Convert predictions to a list (if they are tensors)
val_preds = torch.tensor(val_preds).cpu() # Ensure predictions are also on CPU

# Evaluate accuracy
accuracy = accuracy_score(val_labels, val_preds)
print(f"Validation Accuracy: {accuracy}")
```

Validation Accuracy: 0.8055555555555556

Model Training Code

```

from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report

# Initialize and train the KNN model
model = KNeighborsClassifier(n_neighbors=5) # You can adjust the number of neighbors
model.fit(X_train, y_train)

# Make predictions on the test set
y_pred = model.predict(X_test)

# Evaluate the model
print(f"Accuracy: {accuracy_score(y_test, y_pred):.2f}")
print("Classification Report:")
print(classification_report(y_test, y_pred))

```

Accuracy: 0.86

Classification Report:

	precision	recall	f1-score	support
0	0.85	1.00	0.92	29
1	1.00	0.29	0.44	7
accuracy			0.86	36
macro avg	0.93	0.64	0.68	36
weighted avg	0.88	0.86	0.83	36

Results

BIAS BUSTERS

Detects and mitigates bias in text and images

This tool helps identify potential bias in your content and suggests more inclusive alternatives.

[Text Analysis](#)

[Image Analysis](#)

Analyze text for potential bias

Enter or paste text below to scan it for potential biased language. The tool will identify biased terms and suggest more inclusive alternatives.

[Enter text to analyze](#)

Women are too emotional to handle leadership positions effectively.

Analyze Text for Bias

Example Potentially Biased Text

Analysis Summary

Bias Detected: **Yes**
 Bias Level: **Low**
 Processing Time: 0.10ms

Original Text

Women are too emotional to handle leadership positions effectively.

Detected Issues

Biased Terms Found:

emotional

Bias Categories:

Gender Bias

Suggested Alternative

Women are too passionate to handle leadership positions effectively.

BIAS BUSTERS

Detects and mitigates bias in text and images

This tool helps identify potential bias in your content and suggests more inclusive alternatives.

[Text Analysis](#)

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Analyze text for potential bias

Enter or paste text below to scan it for potential biased language. The tool will identify biased terms and suggest more inclusive alternatives.

[Enter text to analyze](#)

Women and men are equally capable of leadership.

Analyze Text for Bias

Example Potentially Biased Text

Analysis Summary

Bias Detected: **No**
 Bias Level: **None**
 Processing Time: 0.03ms

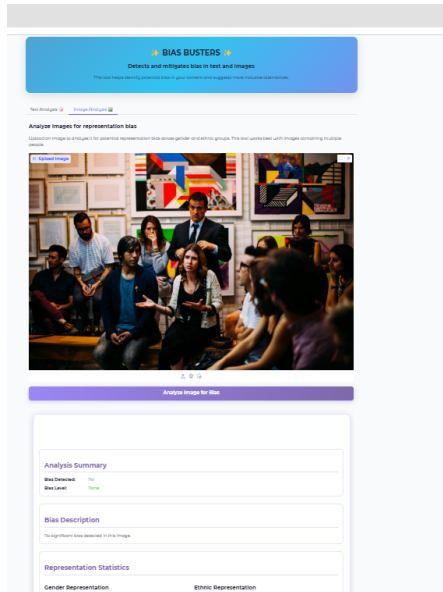
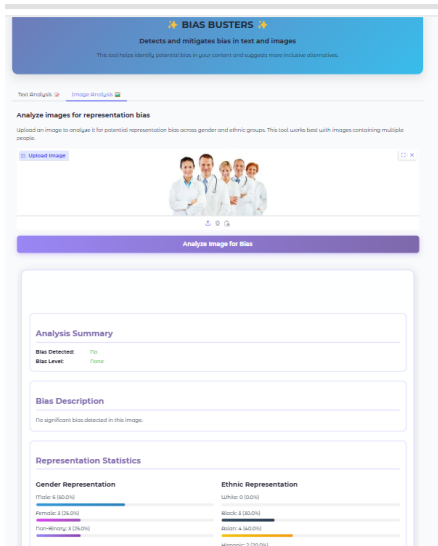
Original Text

Women and men are equally capable of leadership.

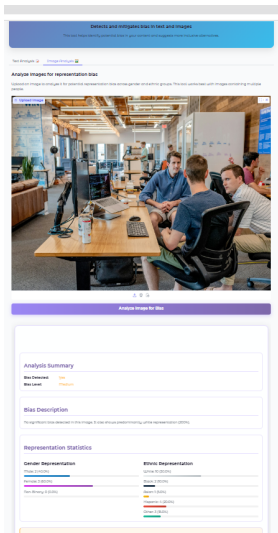
Analysis Details

No bias detected in the provided text. The content appears to be neutral and free from common biased language patterns.

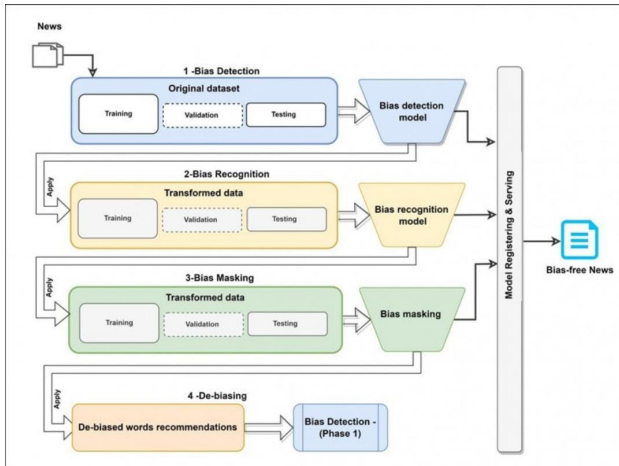
Results



Results



System Architecture



Conclusion

- Successfully built an end-to-end bias detection and mitigation system.
- Created a user-friendly interface accessible to non-technical users.
- Demonstrated effective identification of bias across multiple categories.
- Developed contextually appropriate mitigation strategies.

References

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